

Comparative study of metaheuristic optimised PI controllers for sensorless BLDC motor drives using nonlinear moving horizon estimation

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Abstract

Brushless DC (BLDC) motors are widely preferred due to their high reliability under heavy driving cycles. The factor that makes them reliable which is electrical commutation, can be affected by the hall sensor wearing out over time. To eliminate hall sensors, many sensorless control techniques emerged for speed control of the BLDC motor. This work presents a comparative study of metaheuristic-optimised Proportional-Integral (PI) controllers for a Nonlinear Moving Horizon Estimator (NMHE) based sensorless BLDC drive, integrated with a Hall sensor-assisted startup strategy to overcome the low-speed startup problem. The NMHE is used to estimate rotor position and speed from the measured electrical signals, and the estimated position is then used to generate virtual Hall signals for electronic commutation. To improve speed tracking performance, a comparison of conventional PI and metaheuristic-optimised PI controllers, namely Particle Swarm Optimisation (PSO)-PI, Grey Wolf Optimisation (GWO)-PI, and Dolphin Echolocation Algorithm (DEA)-PI controllers are implemented under the same sensorless motor condition. The comparative performance analysis is validated using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). RMSE of different algorithm approaches is given as 1.647, 0.9538, 0.6075, 0.639 km/hr, respectively. The results demonstrate that the GWO-PI tuning strategy gives the best speed tracking performance. Simulation results indicate that the proposed NMHE-based sensorless structure is stable throughout the drive cycle, while the optimum PI controllers give a better dynamic response than the conventional PI case. Therefore, an NMHE-based sensorless drive with metaheuristic-tuned PI control offers a robust, cost-effective, and high-accuracy solution for EV drive cycle applications.

Introduction

The growth of electric vehicles (EVs) has enhanced sustainable transportation. As the use of EVs continues to increase, the demand for efficient and reliable motor drive systems also increases. Among the various motor drives, Brushless Direct Current (BLDC) motors have gained attention due to their high efficiency, high torque-to-weight ratio, compact size, and low maintenance requirements¹. These benefits of BLDC motors make them

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the best choice for applications in electric vehicles. A major advantage of the BLDC motor is the absence of mechanical commutation. Instead, it uses electrical commutation, which improves its reliability and motor life^{2,3}. However, traditional BLDC motor drives use position sensors such as Hall-effect sensors to determine rotor position for commutation. These sensors increase system cost and reduce reliability under harsh operating conditions⁴. To overcome these limitations, sensorless control techniques are used to estimate the rotor position and speed electrically⁵. Different sensorless control techniques have been developed for BLDC motors. Among these, Back-EMF based methods are widely used for their simplicity. But their performance under low speeds is poor because of weak signal input to the controller^{6,7}. Observer-based methods such as Sliding Mode Observer (SMO) and Extended Kalman Filter (EKF) have been widely used to overcome this limitation⁸. These observer-based methods have higher estimation accuracy, but they also involve a higher computational burden^{9,10}. Moreover, they require accurate parameter adjustment^{11–13, 14}. More advanced techniques, such as particle filter and disturbance observer-based methods, make further improvements to the robustness under nonlinear conditions but at the expense of computational complexity^{15,16}.

Model Predictive Control (MPC) and Moving Horizon Estimator (MHE) are optimisation-based estimation methods that have emerged as effective approaches for handling system nonlinearities and constraints¹⁷. These methods depend on system models and optimisation techniques to provide accurate state estimation under dynamic conditions^{18,19}. Based on this framework, the Nonlinear Moving Horizon Estimator (NMHE) is implemented in this work for sensorless BLDC operation. On the speed control side, Proportional-Integral (PI) controllers are still the most popular BLDC speed control mechanism because of their easy use and simple implementation methods. However, the performance of a PI controller is hugely dependent on proper gain parameter tuning^{20,21}. Traditional PI tuning gives optimal performance under nonlinear and varying operating conditions very rarely. To overcome this limitation, metaheuristic optimisation algorithms such as Particle Swarm Optimisation (PSO) and Grey Wolf Optimisation (GWO) are used to tune the PI gains offline²⁰. The optimised gains are obtained before the drive-cycle simulation and are then held fixed during operation, which avoids the computational burden of online gain adaptation inside the speed loop²². Previous studies show that GWO-based tuning offers superior dynamic response and reduced error over the conventional and PSO-based methods^{23,24}. In the case of EV application, the operating conditions of the motor drive are not constant, and the drive is subject to constant variations of acceleration, deceleration and loads. Under these real drive-cycle conditions, the speed controller and estimation block have to function successfully in an extended speed range²⁵. This necessitates the appropriate choice of a good sensorless estimation method in conjunction with a suitable control strategy to enhance tracking performance and stability of commutation. Hence, sensorless BLDC and optimised PI control become relevant to see how estimation quality, together with controller tuning affects the drive response under realistic load conditions²⁶.

Figure 1 shows the proposed NMHE-based sensorless BLDC drive block diagram that works on a practical drive cycle. The reference speed profile is applied as the input and is compared to the actual motor speed. The error values of reference and motor speed are passed to the outer-loop PI controller. The output of the controller controls the effective DC voltage to the inverter. The inverter drives the BLDC motor using the standard six-step commutation process. At every sampling time, the three phase currents and terminal voltages of the motor are measured. These signals are used to monitor the motor and are also given as inputs to the NMHE-sensorless block. The NMHE uses a nonlinear phase-domain model of the BLDC motor and estimates the electrical position of the rotor. Based on this estimated position, the commutation sector is identified, and virtual Hall signals are generated. These virtual Hall signals are then fed to the same commutation logic in place of physical Hall sensor signals. In this manner, the existing six-step commutation scheme remains unchanged. Another important fact to mention is that sensorless schemes based on back-EMF are known to be difficult at low speeds and during stand-still. This is because the magnitude of the back-EMF depends on the speed and at low speeds the back-EMF is too small to provide reliable position information. It has been suggested in some reviews that sensorless BLDC drives are still challenging during startup, since inaccuracy in the estimation process can cause torque ripple and slow response. Because of this fact, sensorless starting is not considered in the present work. Instead, the Hall sensors

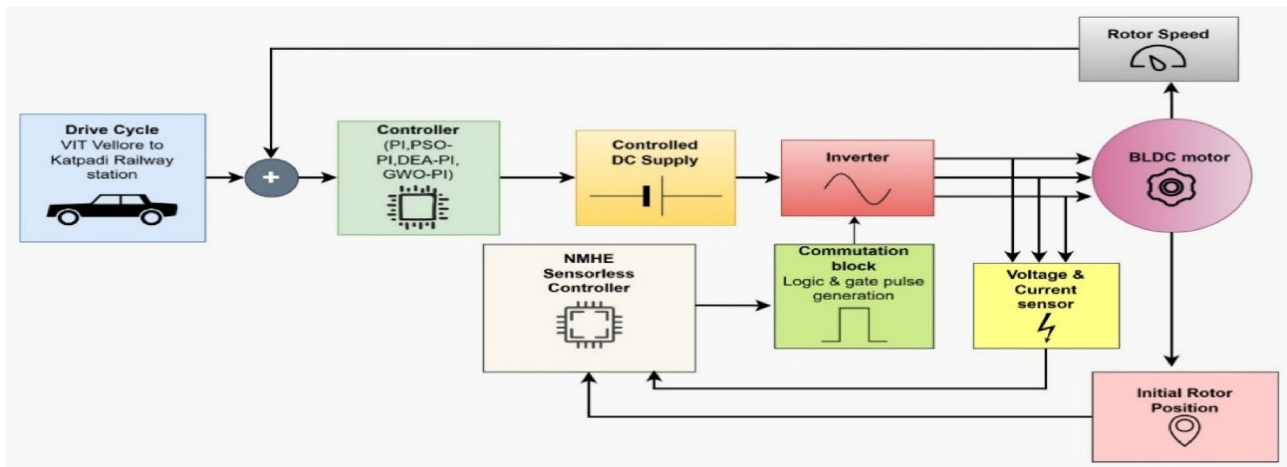


Fig. 1 Block diagram of the NMHE-based sensorless BLDC drive.

are used at the initial stage, when the electrical information available for sensorless estimation is still insufficient. The switch to the NMHE block is made at a given speed; the estimated position replaces the Hall signal and henceforth the drive is switched to full sensorless mode. This hybrid setup is more practical in EV applications where dynamic loading conditions and robust operation over a wide range of speeds are expected.

The main contributions of this work are as follows:

- A comparison of metaheuristic-optimized PI controllers for a sensorless BLDC motor using a Nonlinear Moving Horizon Estimator (NMHE). The NMHE-based estimation technique is developed to estimate rotor position and speed using the measured stator voltages and currents.
- Then, four control approaches, namely Dolphin Echolocation Algorithm (DEA), Grey Wolf Optimization (GWO), Particle Swarm Optimization (PSO), and traditional PI, are implemented and compared using a real-world drive cycle from VIT Vellore to Katpadi Railway Station.
- For performance evaluation of four control approaches, the error-based indices considered in this work are Root Mean Square Error (RMSE) and Mean Absolute Error (MAE).

Literature survey

PID controllers are widely used in the six-step commutation process of speed control of BLDC². However, the effectiveness of the PID controller under dynamic operating conditions is based on the values of the gain parameters^{28,32}. Conventional tuning methods fails to provide optimal values for performance under varying operating conditions^{30,37}. To find optimal solutions in complex and nonlinear systems, Metaheuristic algorithms and Optimization algorithms are used for tuning the parameter values^{31,32}. Even though metaheuristic-based PID controllers exhibit higher computational complexity, they achieve superior accuracy without compromising control performance. Therefore, metaheuristic-based PI controllers are also widely used because of its good balance between control performance and implementation simplicity^{20,24}. From the literature, it is observed that most of the existing works focus on self-tuning PI controller techniques, where gain parameters are adjusted in real time based on feedback to enhance system performance^{33,35,37}. However, these approaches are generally applied to BLDC motors with Hall effect sensors. Integration of dynamically adapted PI control with advanced control sensorless control techniques such as NMHE has not been studied particularly under real-world driving conditions^{17,18}. This motivates the present work, where metaheuristic optimisation techniques are used for offline PI gain tuning in combination with NMHE-based sensorless estimation, and the model performance is evaluated

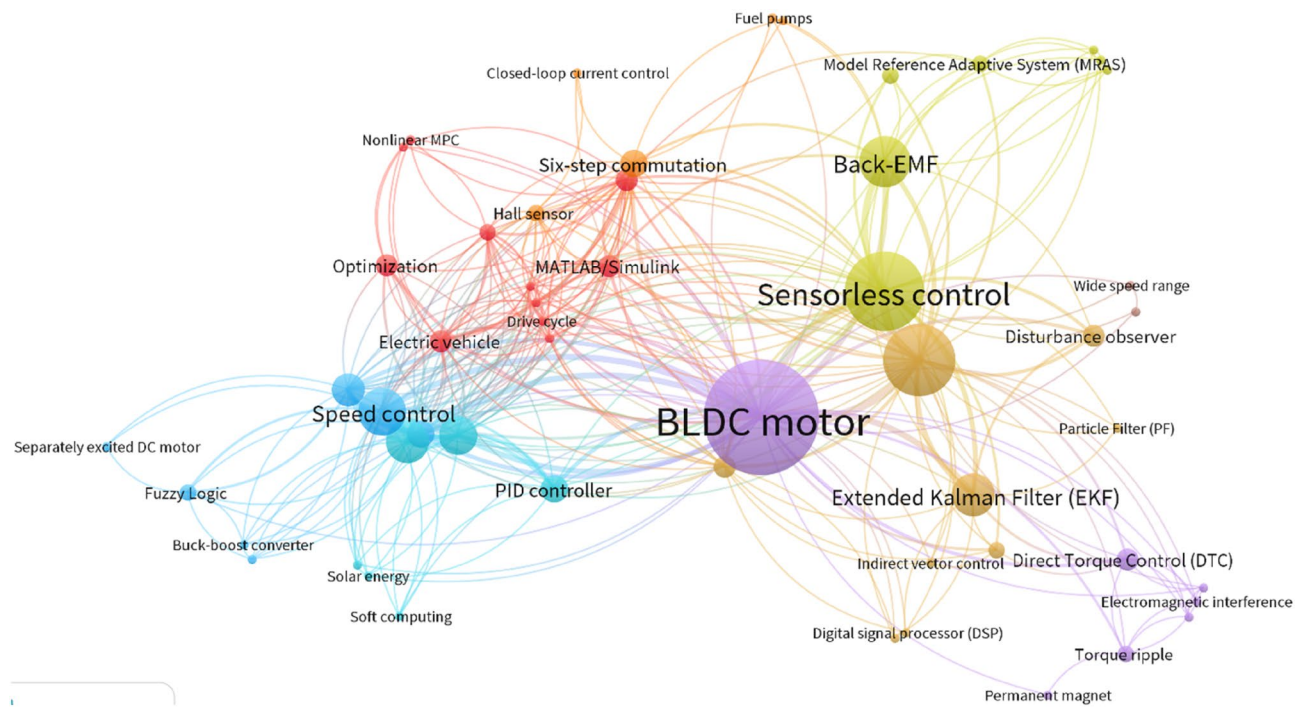


Fig. 2 Connection network of the sensorless BLDC control and advanced controller optimisation literature paper.

Table 2 NMHE Solver and Tuning Settings.

Parameter	Symbol	Value
Horizon length	N	10
Maximum Gauss-Newton iterations	iterMax	6
Convergence tolerance	tol	1×10^{-6}
Levenberg-Marquardt damping	λ	1×10^{-3}
Finite-difference step (speed)	ϵ_{ω}	1×10^{-3}
Finite-difference step (position)	ϵ_{θ}	1×10^{-4}
Sample time	Ts	50 μ s
Speed bound	ω_{max}	6000 rpm
Current clamp	I _{max}	500 A
Output weighting	Q	Identity (no arrival cost)

using a real-world drive cycle. To visually inspect the research linkage between the selected studies, a bibliometric network visualisation was generated from VOSviewer using a key phrase co-occurrence on the selected reference studies which is shown in Fig. 2. There are four clusters with three classes of the two sensorless estimation groups Back-EMF and MRAS^{4,5-6}, Extension Kalman Filter, Particle Filter, and Disturbance observer⁹⁻¹⁵⁻¹⁶, the speed-control group with PID controller, Fuzzy Logic, and Soft computing^{30,35,37}, and the commutation quality group with Direct Torque Control, Torque ripple, and Permanent magnet¹³. This cluster includes Nonlinear MPC, Optimisation, MATLAB/Simulink, Drive cycle, and Electric vehicle. All of these clusters are within the vicinity of sensorless-estimation and speed-control clusters and our study falls into this group¹⁷⁻²³⁻²⁴, but with a lower number of links than is present in EKF or PID. These observations highlight the two shortcomings of this paper: There is no direct comparison of metaheuristically tuned PI controllers with a sensorless BLDC drive via an NMHE, nor does it use a real driving cycle as a reference for that drive.

A critical reading of these works shows two consistent gaps. First, advanced sensorless estimators such as the EKF, particle filter, and disturbance observer are validated mostly under fixed-speed or step references rather.

NMHE-based sensorless estimation

Input: $V_a, V_b, V_c, i_a, i_b, i_c, T_L$
Output: $\omega_{\text{hat}}, \theta_{\text{e_hat}}, H_a_{\text{hat}}, H_b_{\text{hat}}, H_c_{\text{hat}}$

- 1: Update voltage and current buffers over horizon N
- 2: Initialize state vector $z = [\omega, \theta_{\text{e}}]$
- 3: For each iteration do
- 4: Compute residual using BLDC prediction model
- 5: Compute Jacobian and update z
- 6: Apply bounds to speed and position
- 7: End
- 8: Obtain ω_{hat} and $\theta_{\text{e_hat}}$ from optimized state
- 9: Generate virtual Hall signals from $\theta_{\text{e_hat}}$
- 10: Return estimated speed, position, and Hall signals

DEA-Based PI Tuning

Input: speed error e
Output: control signal u
1: Initialize dolphins for K_p and K_i
2: Update integral error
3: For each iteration do
4: Evaluate fitness of each dolphin
5: Find best dolphin
6: Update dolphin positions using exploration and exploitation
7: End for
8: Select best K_p and K_i
9: Compute PI output u

PSO-Based PI Tuning

Input: speed error e
Output: control signal u
1: Initialize particles for K_p and K_i
2: Update integral error
3: For each iteration do
4: fitness of each particle
5: Update personal best and global best
6: Update particle velocity and position
7: End for
8: Select best K_p and K_i

GWO-Based PI Tuning

Input: speed error e
Output: control signal u
1: Initialize wolves for K_p and K_i
2: Update integral error
3: For each iteration do
4: Evaluate fitness of each wolf
5: Select alpha, beta, and delta wolves
6: Update wolf positions using GWO rules
7: End for
8: Select best K_p and K_i
9: Compute PI output u

Algorithm: In the above three column provided the pseudocode for different algorithms. (a) PSO-Based PI Tuning, (b) DEA-Based PI Tuning And (c) GWO-Based PI Tuning.

than under continuously varying real driving profiles, so their position-estimation quality across a full acceleration and deceleration cycle is rarely quantified. Second, metaheuristic PI tuning is almost always demonstrated on Hall-sensored drives, which leaves open whether the same gains remain effective when the speed feedback itself is supplied by an estimator. The specific problem addressed in this work is therefore the joint behaviour of estimator quality and controller tuning within a single sensorless drive under a realistic electric-vehicle speed profile, and how the choice of tuning algorithm, namely PSO, DEA, or GWO, changes the speed-tracking error under these varying operating conditions.

Mathematical model of BLDC motor

BLDC motor is a type of three-phase permanent-magnet synchronous motor that is known for its high efficiency, high torque-to-weight ratio, and reliability^{1,4}. In BLDC drives, the use of permanent magnets replaces the conventional mechanical commutator with electronic commutation, which improves their reliability and lifespan².

The BLDC motor is commonly operated under six-step commutation, and the phase-domain model is generally preferred for analysis, simulation, and controller design^{12,17}. For a symmetrical three-phase BLDC motor, the stator phase-voltage equation for phase $x \in \{a, b, c\}$ can be written in compact form as^{9,12}

$$v_x = Ri_x + L \frac{di_x}{dt} + e_x, x \in \{a, b, c\} \quad (1)$$

where v_x , i_x , and e_x are the phase voltage, phase current, and phase back-EMF, respectively, R is the stator phase resistance, and L is the phase inductance⁹. For a balanced three-phase system, the phase currents satisfy $i_a + i_b + i_c = 0$, which is commonly used to simplify the phase-variable equations¹². The back-EMF of a BLDC motor is trapezoidal in nature and proportional to the rotor electrical position. The three phase back-EMFs are therefore expressed in compact form as^{27,36}

$$e_x = K_e f_x(\theta_e) \omega_m, x \in \{a, b, c\} \quad (2)$$

where K_e is the back-EMF constant, ω_m is the mechanical rotor speed, θ_e is the electrical rotor position, and $f_a(\theta_e)$, $f_b(\theta_e)$, $f_c(\theta_e)$ are trapezoidal back-EMF shape functions shifted by 120° electrical. The trapezoidal back-EMF shape is the main reason for the use of six-step commutation in BLDC motors²⁷. The electromagnetic torque developed by the motor can be expressed either in terms of the instantaneous power balance or in terms of the back-EMF shape functions and phase currents^{2,9}:

$$T_e = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega_m} = K_t [f_a(\theta_e) i_a + f_b(\theta_e) i_b + f_c(\theta_e) i_c] \quad (3)$$

where K_t is the torque constant^{2,9}, $K_t = K_e$ for trapezoidal BLDC with back-EMF constant defined with respect to mechanical speed. In six-step operation, only two phases conduct during each commutation interval, and the third phase remains unexcited^{4,28}. The mechanical dynamics of the motor are given by^{2,12}

$$J \frac{d\omega_m}{dt} = T_e - T_L - B\omega_m \quad (4)$$

where J is the rotor inertia, B is the viscous-friction coefficient, and T_L is the load torque. The electrical rotor position is proportional to the mechanical speed by $\frac{d\theta_e}{dt} = p\omega_m$, where p is the number of pole pairs. Equations (1)–

(4) describe the nonlinear electromechanical dynamics of the BLDC motor used in this work. This nonlinear model forms the base of both sensorless estimation and closed-loop control. Conventional sensorless BLDC drives recover rotor position using back-EMF zero-crossing, terminal-voltage differences, flux-linkage functions, or observer-based estimation methods. More advanced methods based on EKF, particle filtering¹⁵, disturbance observers¹⁶, and optimisation-based estimation such as MPC and MHE have been developed to improve accuracy under nonlinear and dynamic conditions. In the present work, the above BLDC model is used as the prediction model for the Nonlinear Moving Horizon Estimator (NMHE), where rotor position and speed are estimated from the measured stator voltages and currents, and the estimated position is then used to generate virtual Hall signals for six-step commutation.

NMHE-Based sensorless estimation method

In conventional BLDC motor drives, Hall sensors are commonly used to detect rotor position for six-step commutation process. However, it also increases system cost and reduces reliability of the motor, especially under harsh operating conditions. To overcome this problem, sensorless control methods such as back-EMF detection, flux-linkage estimation, sliding mode observer, disturbance observer, EKF, and particle filter methods are developed. Despite this method the algorithms still affected under low-speed, parameter variation, nonlinear and

disturbed load conditions. To address this issue, this work proposes NMHE based sensorless algorithm. NMHE is an optimization-based estimation method that uses the system model and measured signals over a short moving horizon to estimate the internal states more accurately. NMHE is used to estimate the rotor speed and electrical position of the BLDC motor using measured stator voltages and currents. The estimated rotor position is then converted into virtual Hall signals, then fed into commutation logic block. The commutation logic block converts the virtual hall to gate signals. Thus, eliminating the Hall sensors while keeping the commutation structure of the BLDC drive intact.

Estimation states and measured signals

For the proposed sensorless estimation, the state variables are chosen as the rotor mechanical speed ω_m and the electrical rotor position θ_e , so that the estimator state vector is $x = [\omega_m \theta_e]^T$. The measured inputs and outputs fed to the estimator are the three-phase stator voltages and currents, expressed as¹⁷:

$$u = [v_a v_b v_c]^T, y = [i_a i_b i_c]^T \quad (5)$$

NMHE cost function

At each sampling instant, the state estimate is obtained by minimising the prediction error over a moving horizon of length N ^{17,18}:

$$J_k = \sum_{j=k-N+1}^k \|y_j - \hat{y}_j\|_Q^2 \quad (6)$$

where N is the horizon length and the cost term penalises the discrepancy between the measured and model-predicted phase currents across the horizon. The estimator uses identity output weighting ($Q=I$) and no arrival-cost term; regularisation is instead provided by warm-starting the decision variables from the previous estimate together with Levenberg-Marquardt damping in the solver. The cost function is solved using a Gauss-Newton iteration with Levenberg-Marquardt damping. The main computational steps of the NMHE block are summarised in Algorithm 1.

Prediction model and residual

The internal prediction model of the estimator is the phase-domain BLDC model^{14,15}. The terminal voltages are first referred to the floating neutral, with $v_n = \frac{v_a + v_b + v_c}{3}$ and $v_{xn} = v_x - v_n$ for $x \in \{a, b, c\}$. A normalized trapezoidal shape function $F(\theta)$ defines the phase back-EMFs $e_a = K_e \omega F(\theta)$, $e_b = K_e \omega F(\theta - \frac{2\pi}{3})$ and $e_c = K_e \omega F(\theta + \frac{2\pi}{3})$, where the back-EMF constant referred to mechanical speed is $K_e = p\psi$. The continuous model is discretized by forward Euler at the sample time T_s as⁷:

$$i_x[k+1] = i_x[k] + \frac{T_s}{L} (v_{xn}[k] - Ri_x[k] - e_x[k]) \quad (7)$$

$$\omega[k+1] = \max\left(\omega[k] + \frac{T_s}{J} (T_e[k] - T_L - B\omega[k]), 0\right) \quad (8)$$

$$T_e[k] = K_e \left(F(\theta) i_a + F\left(\theta - \frac{2\pi}{3}\right) i_b + F\left(\theta + \frac{2\pi}{3}\right) i_c \right) \quad (9)$$

$$\theta[k+1] = \text{wrap}_{2\pi}(\theta[k] + T_s p \omega[k]) \quad (10)$$

The measured output of the estimator is the three-phase current vector $y[k] = [i_a i_b i_c]$. Over a moving window of N samples, the only decision variables are the mechanical speed and electrical position at the start of the window, $z = [\omega_0 \theta_0]$. The current trajectory is initialized from the first measured sample in the window and propagated by the equations above, so the estimation problem reduces to two unknowns, giving a reduced single-shooting formulation. The residual at step k is the difference between the predicted and measured currents, $r_k(z) = i[k] - i_{\text{meas}}[k]$, and the stacked residual $r(z) = [r_1 \cdots r_N]$ collects all current prediction errors over the horizon. The cost in Eq. (6) is the sum of squares of this residual.

Solver and convergence

Equation (6) is minimized at each sampling instant by Gauss-Newton iteration with Levenberg-Marquardt damping. The Jacobian $J_G = \partial r / \partial z$ is evaluated by forward finite differences with steps ϵ_ω and ϵ_θ ¹²:

$$J_G[:, 1] = \frac{r(\omega_0 + \epsilon_\omega, \theta_0) - r(z)}{\epsilon_\omega}, J_G[:, 2] = \frac{r(\omega_0, \theta_0 + \epsilon_\theta) - r(z)}{\epsilon_\theta} \quad (11)$$

and each iteration solves the damped normal Eq.

$$(J_G^T J_G + \lambda I) \Delta z = -J_G^T r(z), z \leftarrow z + \Delta z \quad (12)$$

After every update the speed is projected to the bound $0 \leq \omega \leq \omega_{\max}$ and the position is wrapped to $[0, 2\pi)$. The iteration stops when the update norm satisfies $\|\Delta z\| < \text{tol}$ or after iterMax iterations. The converged z is then propagated to the end of the window to obtain the present speed and position estimates, which also warm-start the next sampling instant; a current clamp of $\pm I_{\max}$ is applied during this end-of-window propagation. The solver and tuning settings used in this work are listed in Table 1.

The estimated speed and electrical position obtained from Algorithm 1 are then used for sensorless commutation and feedback control. The system is therefore Hall-assisted sensorless, not fully sensorless. Hall sensors are used only during startup. The transition condition is defined as a fixed threshold on the measured mechanical speed, ω_{switch} is 300 rpm. Once $\omega \geq \omega_{\text{switch}}$ is sustained for 50 ms to avoid chattering near the threshold, the commutation logic switches from physical Hall signals to the NMHE-derived virtual Hall signals, and Hall sensor inputs are no longer used.

Implementation of the proposed NMHE

The NMHE algorithm is realized in MATLAB/Simulink with a moving horizon value of ten. The state values at the end of the horizon are taken as the present estimated values. The estimator is initialized on startup using Hall-assisted BLDC operation, so that the motor can start with stable commutation. When the motor reaches appropriate working conditions, the hall assist is turned off and NMHE block is used to estimate rotor position and speed for the sensorless operation. The estimator can't start the BLDC by itself, so it takes over the motor when it is already assisted in the commutation process. This makes the proposed approach practical and stable for simulation-based BLDC drive implementation.

Virtual hall signal generation

After estimating the rotor electrical position θ_e , virtual Hall signals are generated for six-step commutation by examining the electrical position over the 60° commutation sectors. The three virtual Hall signals are given in compact form as²⁰:

$$H_k = H(\sin(\theta_e + \phi_k)), \phi_k \in \left\{0, -\frac{2\pi}{3}, +\frac{2\pi}{3}\right\}, k \in \{a, b, c\} \quad (13)$$

Table 1 summarises the different PI controller tuning and optimisation techniques, along with their advantages and limitations from the reference papers.

Ref.	Method Category	Technique	Methodology	Advantages	Limitations
²	Conventional control	Closed-loop PI control	Fixed-gain PI controller used in closed-loop BLDC speed regulation	Simple and widely used	Poor performance under nonlinear conditions
⁴	Sensorless control	Review of sensorless methodS	Comparative study of Back-EMF, observers, AI methodS	Broad understanding of techniqueS	No implementation or validation
⁶	Sensorless control	Back-EMF detection	Zero-crossing based rotor position estimation	Simple, low cost	Poor performance at low speed
⁹	Estimation technique	Extended Kalman Filter (EKF)	State estimation using non-linear Kalman filtering	High estimation accuracy	High computational burden'
¹¹	Estimation technique	EKF with disturbance handling	EKF with compensation for disturbances	Improved robustness	Complex implementation
¹⁶	Observer-based control	Disturbance observer	Estimation of unknown disturbances in BLDC	Improved robustness	Limited adaptability
¹⁷	Optimization-based control	Model Predictive Control (MPC)	Optimization-based control using system model	Handles constraints effectively	Requires accurate model
²⁰	Metaheuristic control	PSO & GWO-based PI tuning	PI gains optimized using PSO and GWO based objective-function minimization	Improved transient response, GWO better than PSO	Offline optimization focus
^{23,24}	Metaheuristic control	GWO-based PI/PID tuning	Controller gains tuned using Grey Wolf Optimization search strategy	Better convergence, improved dynamic response	Requires parameter selection
³⁰	Metaheuristic control	PSO-based PID tuning	PID gains obtained using PSO-based optimization	Simple implementation, good global search	May converge to local optimum
³³	Metaheuristic control	PSO-optimized BLDC control	BLDC speed control enhanced through PSO-based parameter optimization	Improves speed response	Limited to PSO
^{34, 35}	Metaheuristic control	PSO & TLBO-based PID tuning	PID parameters tuned using hybrid optimization schemes based on PSO and TLBO	Better optimization flexibility	Increased computational effort
^{31,32}	Optimization theory	GWO & PSO algorithms	Population-based search algorithms used as general-purpose optimizers	Strong global optimization capability	Not application-specific
^{37,38}	Intelligent control	Fuzzy-based self-tuning PI	PI gains adjusted online using fuzzy logic rules based on system error/response	Adaptive gain adjustment	Complex design, rule dependency

where the switching function is defined as $H(x) = 1$ for $x \geq 0$ and $H(x) = 0$ otherwise. These virtual Hall outputs are then supplied to the commutation logic block to produce the gate pulses for the inverter.

Role of NMHE in the proposed system

In the proposed BLDC drive, the NMHE block performs the following functions:

1. estimates rotor speed and electrical position from measured voltages and currents,
2. replaces the physical Hall sensors during sensorless operation,

3. generates virtual Hall signals for six-step commutation, and.
4. provides the estimated speed signal for the outer-loop PI speed controller.

By combining NMHE-based sensorless estimation with metaheuristic-optimized PI controllers, the proposed work evaluates both estimation and control performance under a real-world drive cycle.

Metaheuristic-optimized PI control strategy

The speed data obtained from the real-world drive cycle is compared with the measured motor speed of the motor. The difference between them is processed by the PI controller, and the controller output drives the BLDC drive under six-step commutation. The PI speed controller is expressed as^{16,22}

$$u(t) = K_p e(t) + K_i \int e(t) dt \quad (8)$$

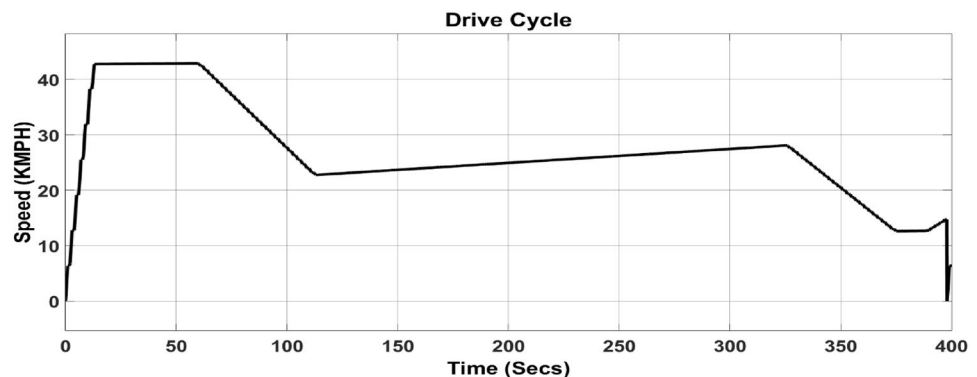
where $e(t)$ is the speed error, and K_p and K_i are the proportional and integral gains, respectively. Different gain selection strategies are considered to study their effect on drive-cycle tracking performance. To improve the gain selection of the PI controller, three metaheuristic optimisation methods, Particle Swarm Optimisation (PSO), Dolphin Echolocation Algorithm (DEA), and Grey Wolf Optimisation (GWO), are employed. These algorithms are used to determine improved values of K_p and K_i for the same NMHE-based sensorless BLDC drive. Their main steps are summarized in Algorithms 2–4.

Real-world drive cycle data

The reference speed profile for this work is acquired from real-world drive cycle from VIT Vellore University to Katpadi Railway Station by using Google Directions API, Google Elevation API and OpenWeatherMap API³⁹. Figure 3. shows the Real-world drive cycle obtained. The profile has acceleration, constant-speed section, deceleration, and low-speed sections. Therefore, this has been used to test the tracking performance of the NMHE based sensorless BLDC drive under practical operating conditions. The route selected is from VIT Vellore University to Katpadi Railway Station, illustrated in Fig. 4. Steps of making the specified experimental route:

- Google direction API is used to fetch direction and rout information between the initial and final positions. from this, information like travelled distance, time travelled can be deduced.

Fig. 3 Real-world drive cycle used as the speed reference input.



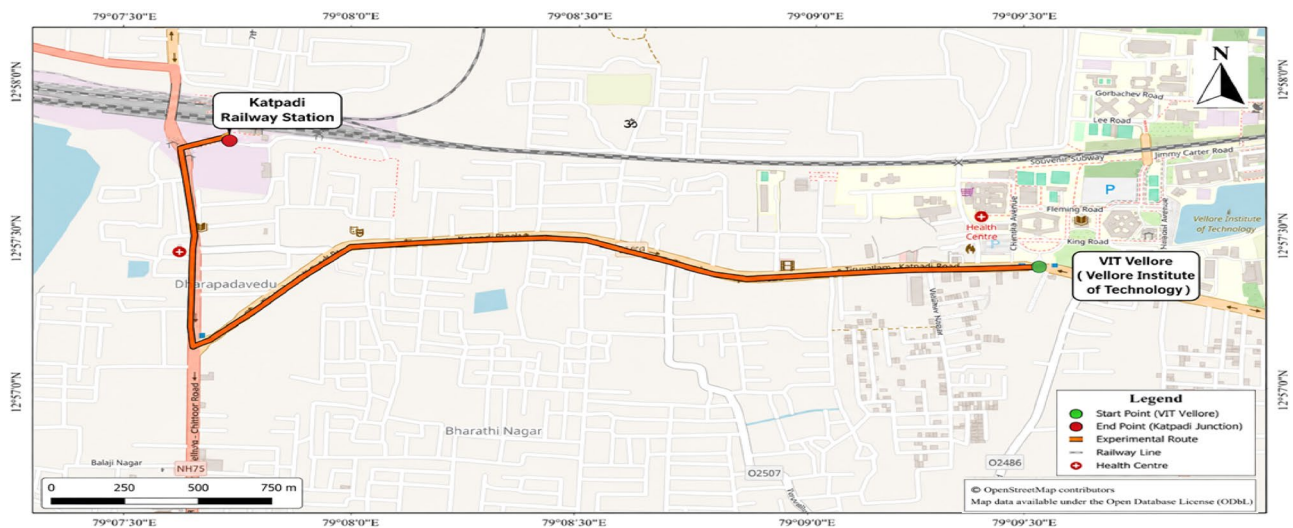


Fig. 4 Map of the experimental route from Vellore Institute of Technology (VIT), Vellore to Katpadi Junction, Vellore. The map was generated by the authors using QGIS(<https://qgis.org/>) with OpenStreetMap data. Map data © OpenStreetMap contributors, available under the Open Database License (ODbL).

Table 3 Electric Vehicle Parameters Used in Simulation.

Parameter	Rating	Unit
No. of wheels per axle	1	-
Frontal area	0.6	m ²
Wheel radius	0.25	m
Gear reduction Ratio	8:1	-
Drag Coefficient	0.24	-
Rolling resistance coefficient	0.015	-
Acceleration due to gravity	9.81	m/s ²
Air density	1.18	Kg/m ³
Ambient temperature	30	°c
Range	5.92	km

- Google Elevation API gives the information about the inclination of road. Data like inclination, declination of roads in the rout travelled by the vehicle.
- OpenWeatherMap API gives the weather-related data at the EV’s location. Provide information such are wind speed and atmospheric pressure.

The estimated travel time and distance between each successive pair of waypoints under typical traffic conditions are provided when the Google Directions API delivers the route polyline. The raw API answer automatically includes a waiting-time component in the segment travel time at waypoints that correspond to signalised junctions in order to represent the average signal delay. Each inter-waypoint travel time was adjusted by deducting the anticipated signal cycle delay at that intersection, which was calculated from the average green-to-red cycle duration derived from Google Maps live traffic data for each intersection along the route, to separate the pure riding time from signal dwell. The very same finite difference equation employed in the stop-and-go cycle was applied to compute for the instant velocity at every single waypoint by employing the time stamps that have already been corrected after removing the dwell times. The output is a speed graph wherein the automobile gradually slows down when nearing a junction but will not come to a full stop and will continue at a reduced speed, which corresponds to the riding condition where there is a green light at every traffic signal. Thus, the computed velocity graph is the best-case real-life riding condition on this corridor where no time is spent idling. Road gradient and

environmental data obtained via the Google Elevation and OpenWeatherMap APIs are kept unchanged compared to the data used during the stop-and-go cycle, because the real geometries of the road and environmental factors are independent from the operation of signals..

Vehicle-to-motor speed conversion

The reference profile obtained in Sect. 6 is expressed in vehicle road speed v (km/h) and must be converted to a motor reference. The wheel's angular speed is velocity divided by the wheel radius. The motor speed is related to the wheel speed through the fixed gear ratio G of the drivetrain, $\omega_{\text{motor}} = \text{Gear ratio} \times \text{The wheel's angular speed}$. The load torque reflected to the motor shaft follows the standard road-load equation. This table III provides the essential aerodynamic, geometric, and environmental parameters required for electric vehicle performance modelling Table 3.

Results and discussion

The NMHE based sensorless BLDC drive is tested using the real-world drive cycle which starts from VIT vellore and ends at Katpadi railway station. The controller comparison is done for conventional PI, PSO-PI, DEA-PI, and GWO-PI. The comparison is made from the drive cycle response, speed tracking behaviour, RMSE and MAE plots, internal waveforms, and the final error indices. Below the simulation setup, error and waveform comparison are discussed.

MATLAB simulation setup

Then, PSO, GWO, and DEA are compared with the conventional PI controller. All four controllers are tested using the same NMHE-based sensorless BLDC drive and the same real-world drive cycle, so that the performance variation is focused only to the gain selection method. The developed BLDC drive system is implemented in MATLAB/Simulink using the mathematical model, NMHE-based sensorless estimation block, and PI-based outer speed controller described in the previous sections. The same BLDC motor model is used for all controller comparisons so that the influence of the gain selection strategy can be studied under identical operating conditions. The motor parameters used in the simulation are shown in Table 4.

During startup, Hall-assisted commutation is used to ensure stable initial motor operation. After reaching suitable operating conditions, the Hall assistance is removed, and the NMHE block provides the estimated rotor speed and electrical position for fully sensorless operation. The estimated electrical position is converted into virtual Hall signals, which are then used for six-step commutation. This arrangement allows the sensorless estimation block and the speed controller to operate together in the same BLDC drive structure. To evaluate the

Table 4 Motor Parameters.

Parameter	Symbol	Value
Phase resistance	R	$1.43 \, \Omega$
Phase inductance	L	$9.4 \times 10^{-3} \, \text{H}$
Flux linkage	ψ	$0.2158 \, \text{Wb}$
Rotor inertia	J	$5.5 \times 10^{-3} \, \text{kg.m}^2$
Viscous friction coefficient	B	$2 \times 10^{-3} \, \text{N.m.s/rad}$
Pole pairs	p	2
Sample time	T_s	$5 \times 10^{-5} \, \text{s}$
NMHE horizon length	N	10
Drive cycle duration	T	400 s
Load Torque	TL	10 N.m

effect of controller, gain selection, four speed control strategies are considered: conventional PI, PSO-based PI, DEA-based PI, and GWO-based PI. Each controller is tested using the same NMHE-based sensorless BLDC drive under the same reference speed profile. The reference speed is taken from a real-world drive cycle corresponding to the route from VIT Vellore to Katpadi Railway Station. The total drive-cycle duration is 400 s and the maximum reference speed is 43 km/h. Therefore, the controller comparison is carried out under realistic and continuously varying operating conditions rather than under fixed-speed test conditions. A constant load torque of 10 N·m is used as a nominal average load so that the four controllers are compared under identical, controlled conditions, isolating the effect of the gain-tuning method from load variation. A time-varying load derived from the road-load model of Sect. 6.1 was additionally applied to the best-performing controller (GWO-PI) to verify robustness to a more realistic load profile.

Performance metrics

MAE and RMSE were employed to evaluate the speed-tracking performance of the proposed electric vehicle model by comparing the reference drive cycle speed with the simulated vehicle speed. Lower values of these metrics indicate better agreement between the desired and actual vehicle responses, reflecting improved tracking accuracy and controller performance. Root Mean Square Error (RMSE) measures the square root of the average squared difference between the reference drive cycle and the actual vehicle output. Mean Absolute Error (MAE) measures the average magnitude of the error between the reference drive cycle and the actual vehicle output without considering the direction of the error. The RMSE value is given by:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N (V_{ref,i} - V_{act,i})^2} \quad (14)$$

$$\text{MAE} = \frac{1}{N} \sum_{k=1}^N |V_{ref,i} - V_{act,i}| \quad (15)$$

V_{ref} is Reference vehicle speed from the drive cycle, V_{act} is the actual vehicle speed, N is the Total number of samples.

Speed tracking analysis

Figure 5 shows Speed responses of different algorithms. It is clearly visible that the optimized PI controllers perform better than the conventional PI controller. PSO-PI gives better response than Conventional PI, DEA-PI provides smoother driving. And overall GWO-PI gives the accurate response to the reference and the lowest overall error. It also gives a smoother response during speed transitions.

Internal waveform validation

To validate the operation of the NMHE-based sensorless BLDC drive, the internal waveforms of all four controllers are observed. The waveforms shown are the three-phase back-EMF, inverter output voltages, developed torque, and virtual Hall signals. These waveforms will be used to determine the commutation works over the entire drive cycle and whether the estimated rotor position provides the right switching sequence for sensorless operation.

Back-EMF signal analysis

The three-phase back-EMF waveforms obtained for all four controllers are shown in Figure. 6. The waveforms remain balanced and follow the variation in motor speed over the drive cycle. The waveform magnitude changes

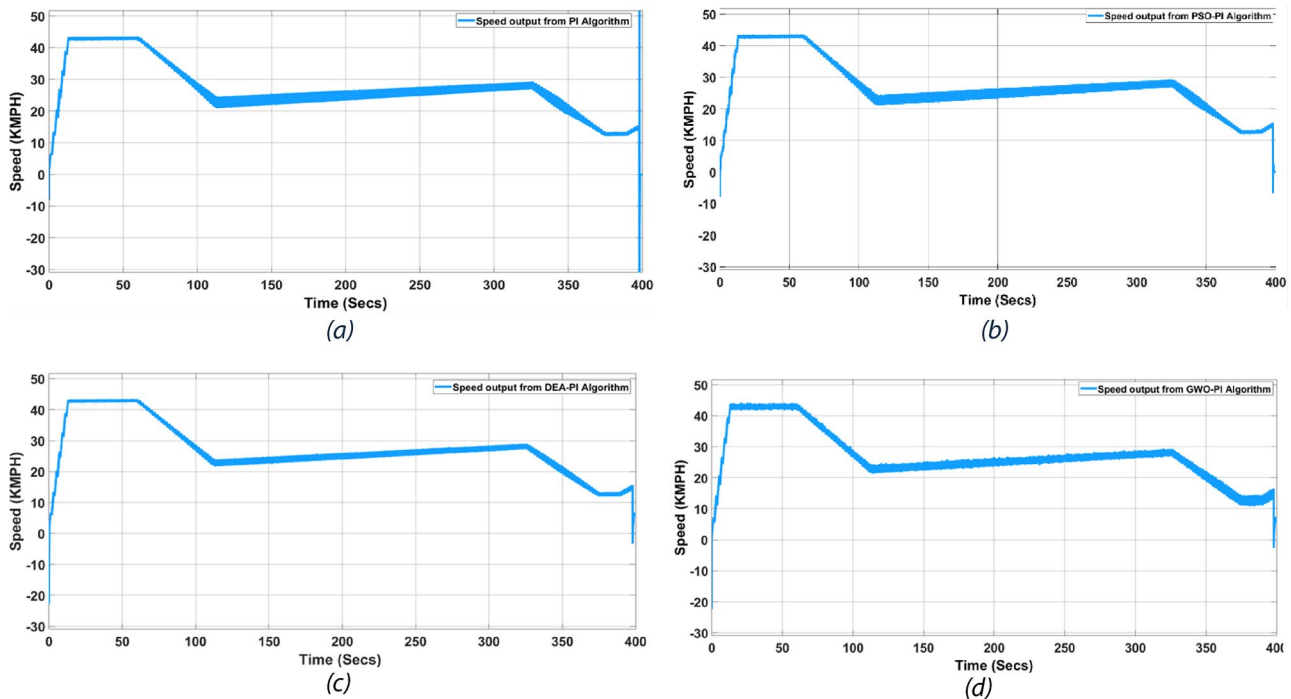


Fig. 5 Speed responses of conventional (a) PI, (b) PSO-PI, (c) DEA-PI, and (d) GWO-PI controllers for the real-world drive cycle.

proportionally with the operating speed, which is consistent with the expected BLDC motor behaviour described by Eq. (2). Across all controllers, the trapezoidal shape of the back-EMF is similar, confirming accurate working of sensorless commutation throughout the drive cycle.

Inverter voltage signal analysis

The inverter output voltages vary according to the switching states required for six-step commutation. The voltage level changes with the controller action and operating condition of the motor. The waveforms of the inverter output voltage are shown in Fig. 7.

Electromagnetic torque signal analysis

The developed electromagnetic torque waveforms of the four controllers are shown in Figure. 8. The torque waveforms are observed to be stable. Hence confirming the stable operation of the metaheuristic-optimized PI controllers. The small torque variations are visible for short time because of rapid speed transitions, which correspond to the acceleration and deceleration phases of the drive cycle. It is observed that conventional PI controller becomes unstable during a very large transition of speed but metaheuristic-optimized PI controllers works well because of its ability to self-adjust gain values.

Virtual hall signal analysis

The virtual Hall signals generated from the NMHE-estimated rotor position are shown in Figure. 9 for all four controllers. The signal represents the six-step switching sequence. The Hall signal transitions are accurate and smooth across all operating speeds, which validates the effectiveness of the NMHE-based position estimation algorithm described in Section IV.

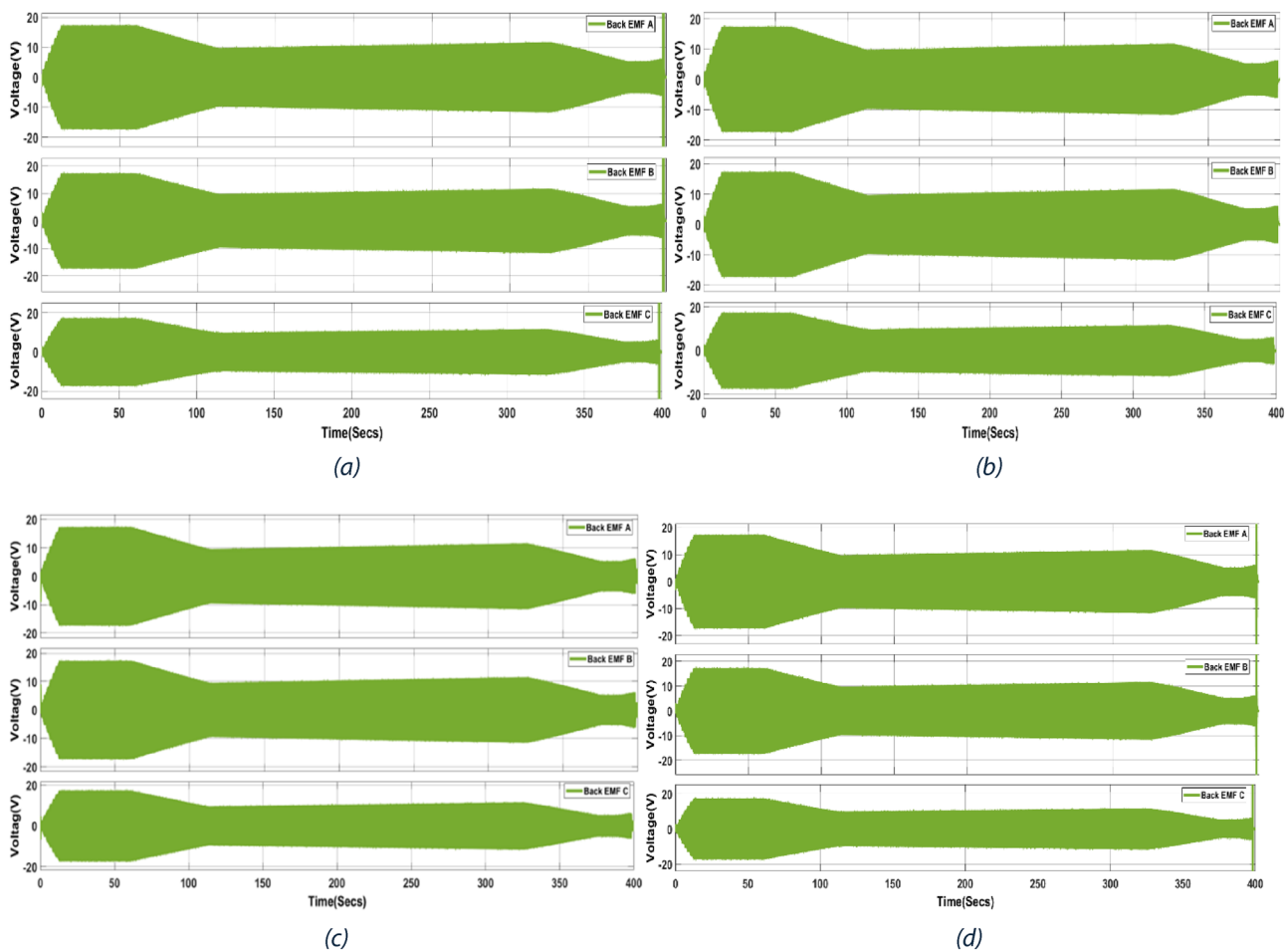


Fig. 6 Three-phase back-EMF waveforms under the real-world drive cycle: (a) Conventional PI, (b) PSO-PI, (c) DEA-PI, and (d) GWO-PI.

Comparative error analysis

In order to compare the four controllers in terms of error performance, the variation of RMSE and MAE during the drive cycle is simulated and tabulated. These plots indicate the variation of the tracking error versus time of conventional PI, PSO-PI, DEA-PI, and GWO-PI controllers during the complete 400 s drive cycle. As RMSE and MAE are derived based on the instantaneous error in speed, the resulting responses provide a more detailed picture of how fast the error will decrease and how all the controllers will track in various operating conditions. Therefore, along with the final values of the error, the RMSE, and MAE graphs can be used to learn about the time-dependent error behaviour of the controllers when driving Figs. 10 and 11.

Among the optimized PI controllers, GWO-PI yields the overall lowest error response, and DEA-PI also demonstrates lower error responses than others. The PSO-PI gives an improved response compared with the conventional PI case though its overall performance is lower than DEA-PI and GWO-PI. Table 5 shows the final error values obtained with the four controllers. The results of the error analysis are tabulated in Table 5.

From the obtained results, the optimized PI controllers give better performance than the conventional PI controller in the main error indices. This shows that improving the PI gains improves the speed tracking performance of the sensorless BLDC drive. Among the optimized methods, GWO gives the lowest overall error and the best tracking response, which also agrees with the trend reported in ^{20,23,24}. The improvement obtained with the optimized controllers is mainly due to better gain selection for the outer speed loop. Since the reference speed varies

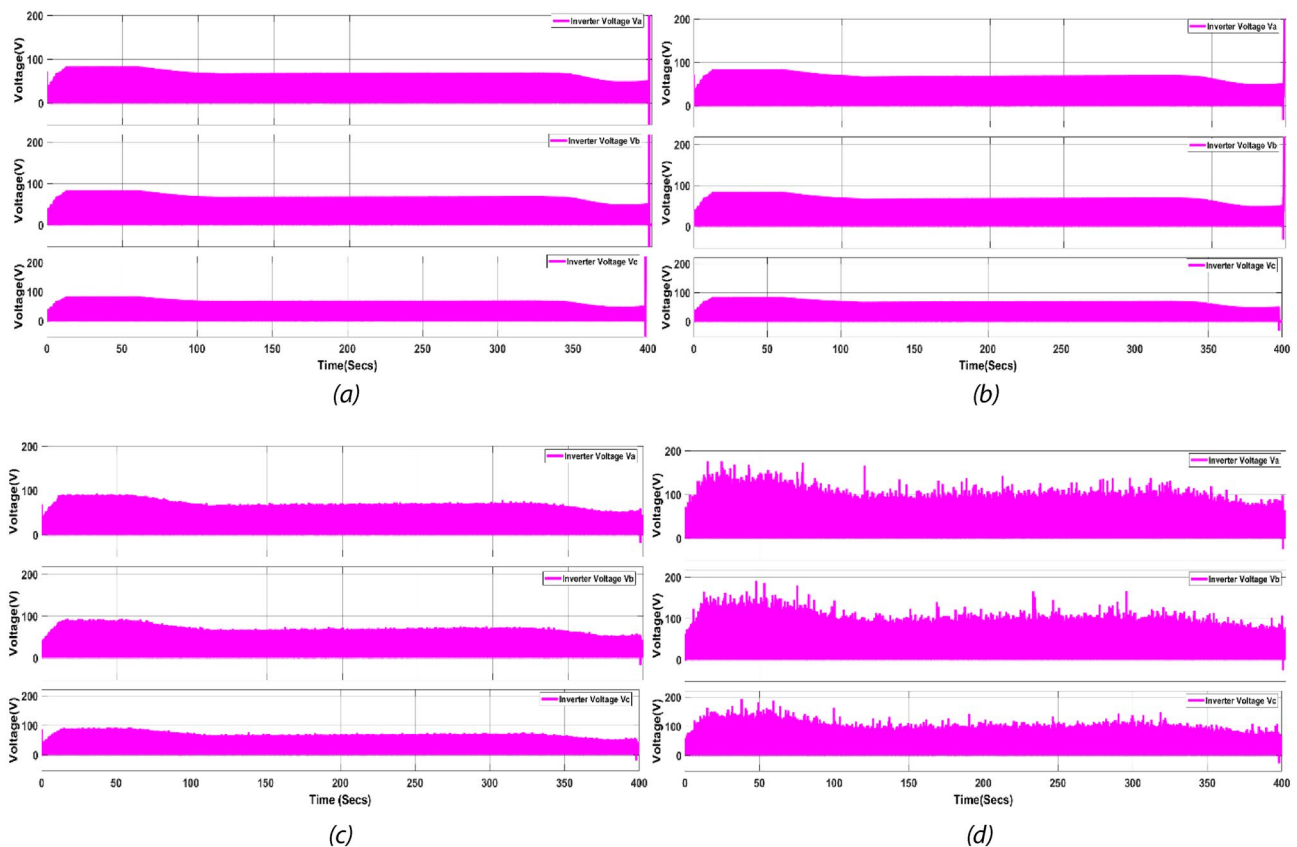


Fig. 7 Three-phase inverter output voltages under the real-world drive cycle: (a) Conventional PI, (b) PSO-PI, (c) DEA-PI, and (d) GWO-PI.

continuously in the real-world drive cycle, fixed conventional PI gains are not always suitable for all operating regions. Metaheuristic tuning improves the balance between transient response and steady tracking performance. In addition, the NMHE block supports the controller by providing continuous speed and position estimation for sensorless operation, which allows the BLDC drive to operate without physical Hall sensors after startup. The GWO-based PI controller gives the best overall result because it provides more effective gain adaptation for the speed loop, resulting in lower tracking error and improved response during dynamic changes in the drive cycle. PSO and DEA also improve the tracking performance compared to the conventional PI controller, but their overall results remain slightly inferior to those of GWO under the tested conditions.

Conclusion and future scope

This work developed an NMHE-based sensorless BLDC drive with Hall-assisted startup and compared four PI tuning approaches, namely conventional PI, PSO-PI, DEA-PI and GWO-PI, under a real-world GPS-derived speed profile from VIT Vellore to Katpadi Railway Station. In simulation, the NMHE block provided continuous speed and electrical-position estimates sufficient to generate virtual Hall signals and sustain six-step commutation across the 400 s profile for all four controllers. The back-EMF, virtual Hall, torque and inverter voltage waveforms remained stable over the entire drive cycle. Among the tuning methods, GWO-PI gave the lowest RMSE and MAE, followed by DEA-PI and PSO-PI, while conventional PI showed the largest tracking deviation. The relative advantage of GWO-PI is attributed to its balance between exploration and exploitation, which yields effective gain selection under continuously varying speed. These results are based on simulation under nominal

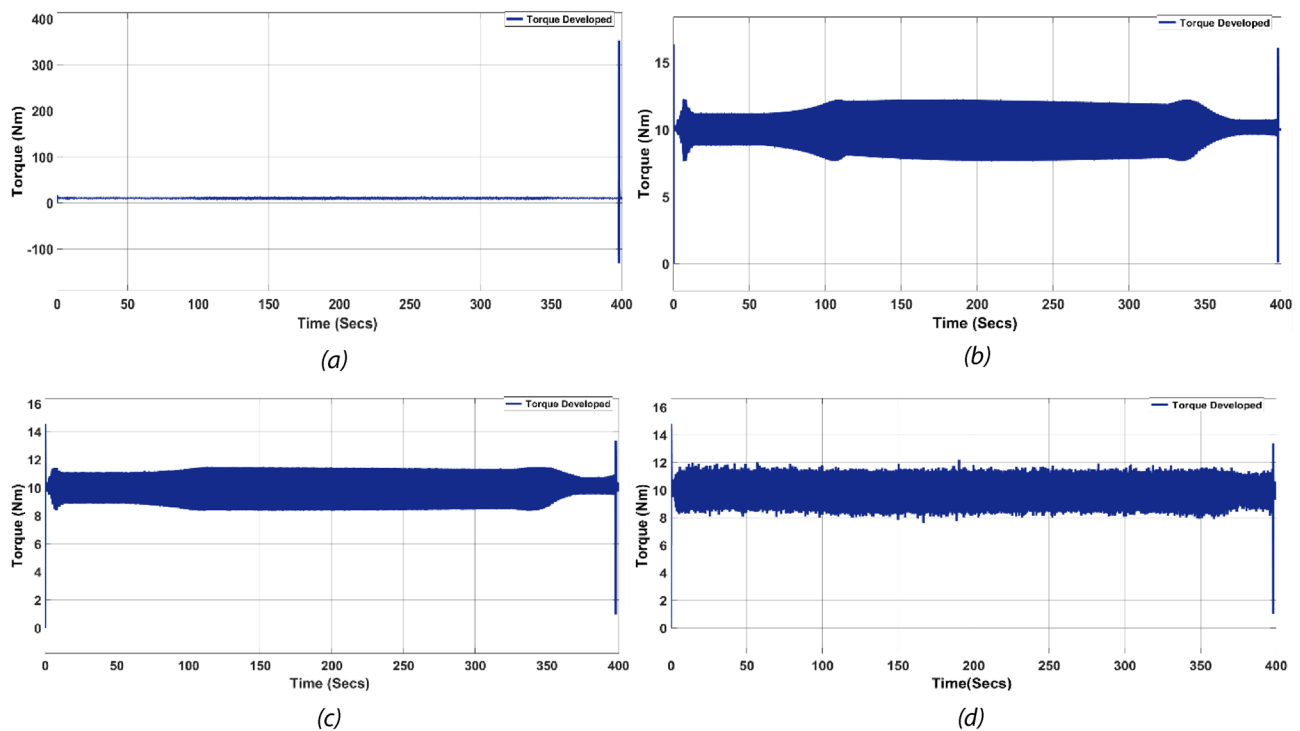


Fig. 8 Developed electromagnetic torque under the real-world drive cycle: (a) Conventional PI, (b) PSO-PI, (c) DEA-PI, and (d) GWO-PI.

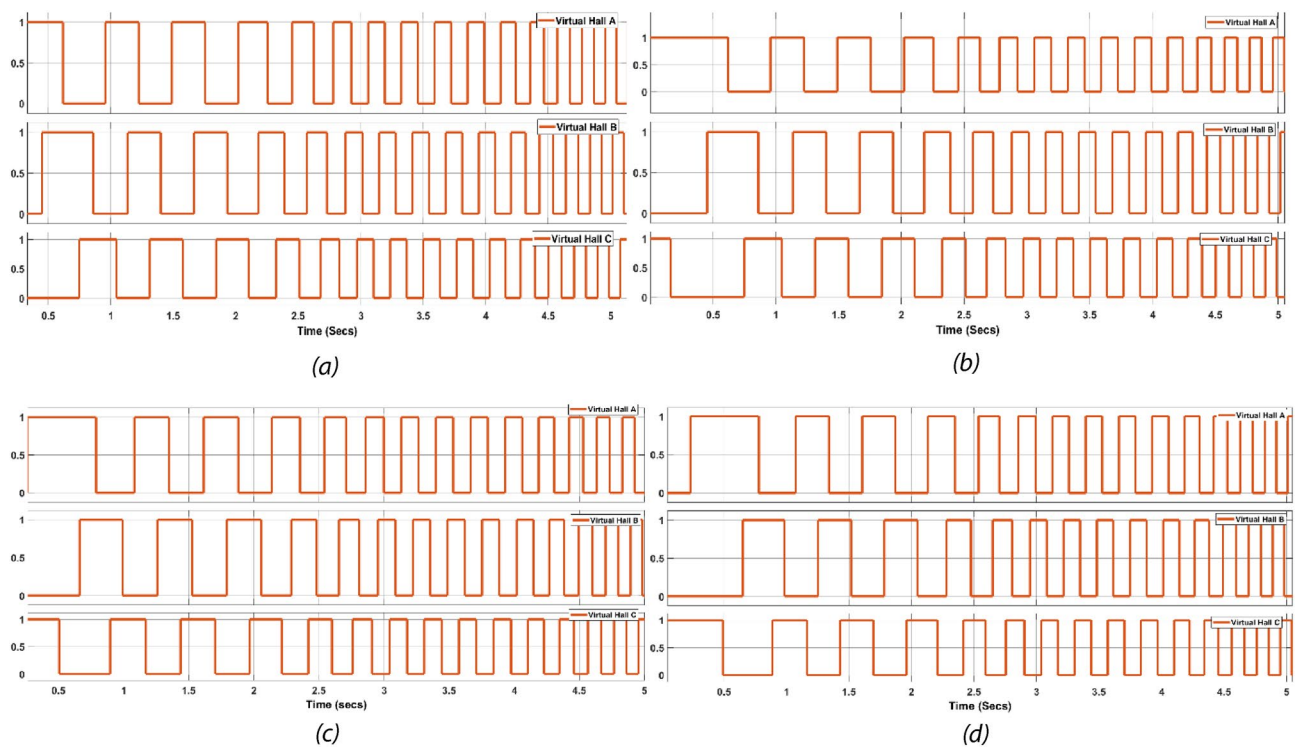


Fig. 9 Zoomed virtual Hall signals generated from the NMHE-estimated rotor position: (a) Conventional PI, (b) PSO-PI, (c) DEA-PI, and (d) GWO-PI.

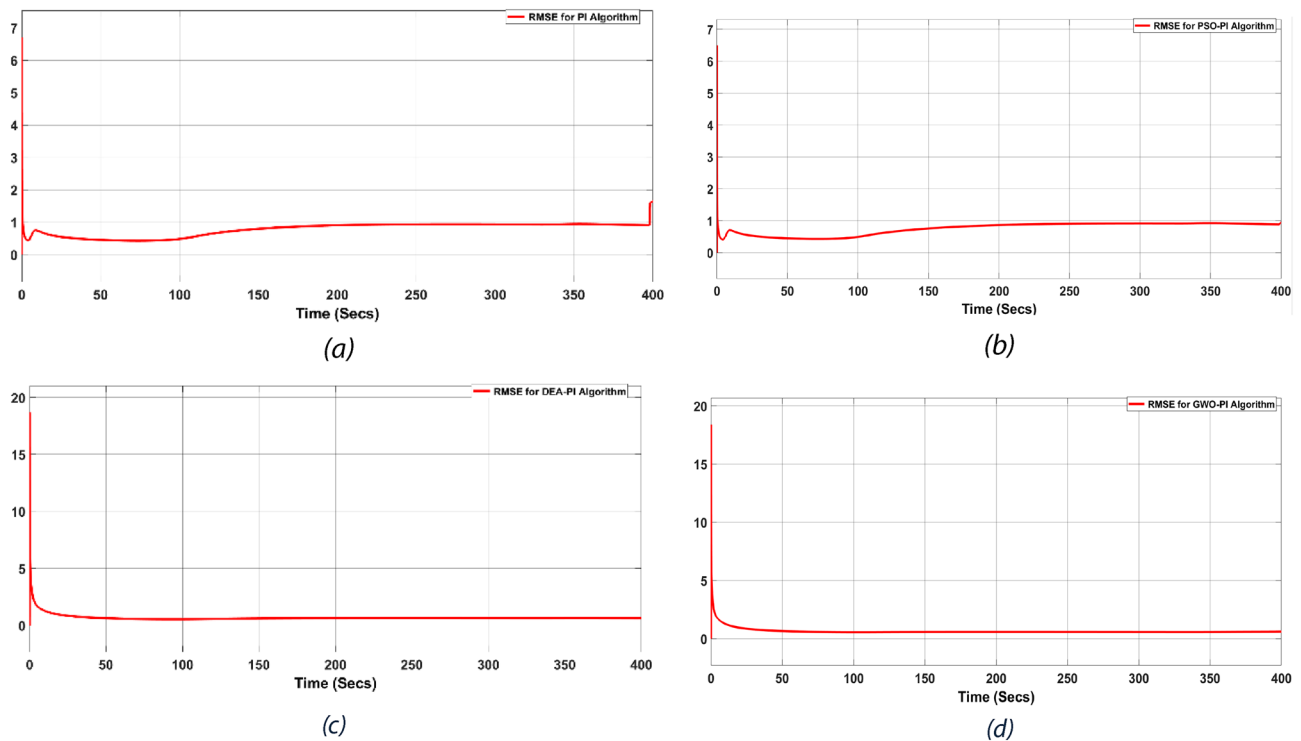


Fig. 10 RMSE of analysis of (a) Conventional PI, (b) PSO-PI, (c) DEA-PI, and (d) GWO-P with respect to reference drive cycle.

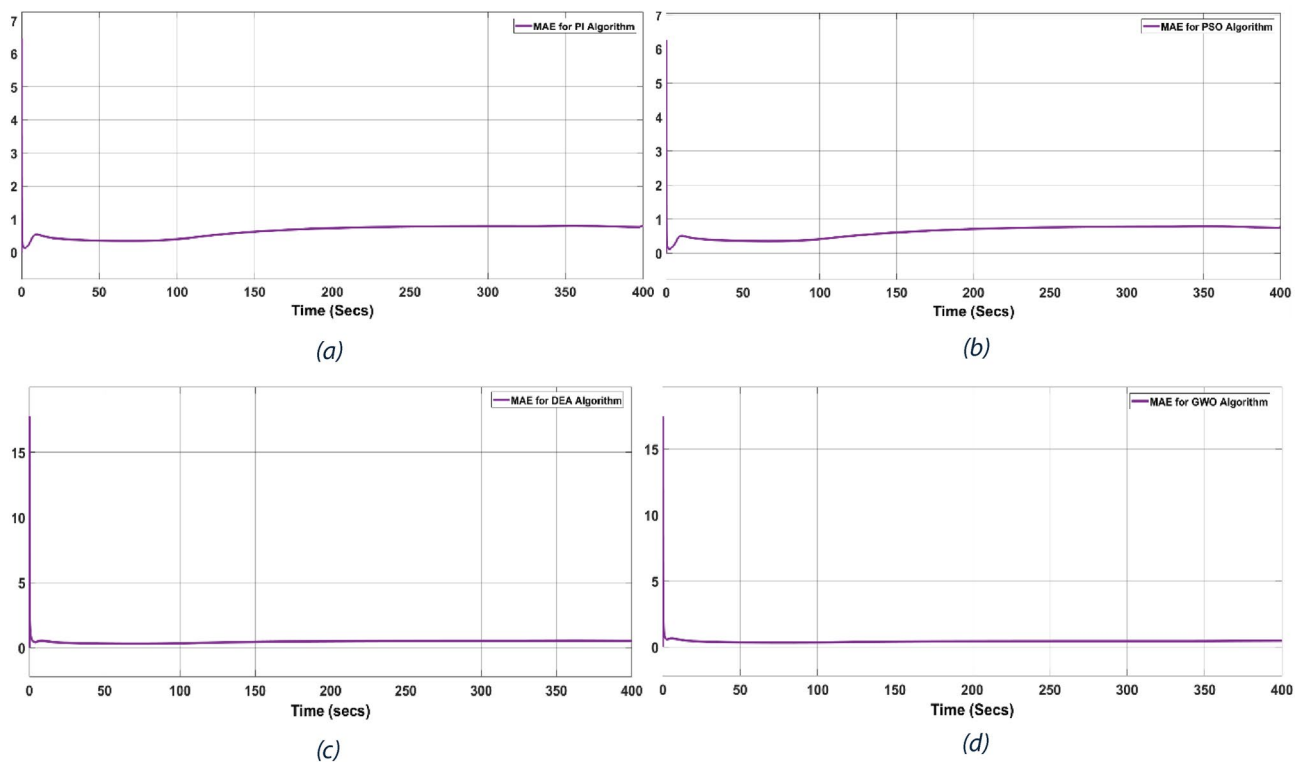


Fig. 11 MAE of analysis of (a) Conventional PI, (b) PSO-PI, (c) DEA-PI, and (d) GWO-P with respect to reference drive cycle.

Table 5 Error analysis of different methods.

Controller	RMSE (km/hr)	MAE (km/hr)	Observation
Conventional PI	1.647	0.8052	Largest tracking deviation
PSO-PI	0.9538	0.7630	Improved over conventional PI, but weaker overall than DEA and GWO
DEA-PI	0.639	0.5298	Better adaptation under varying conditions
GWO-PI	0.6075	0.4876	Best overall performance

parameters; estimator accuracy against ground truth, robustness to parameter mismatch and measurement noise, and real-time computational feasibility were not experimentally verified and are identified as the main limitations of the present study.

Future research can focus on real-time implementation of the NMHE algorithm in hardware. For, hardware implementation, reducing the horizon length or using warm-starting techniques can bring down computation time. Model analysis under motor parameter variations such as thermal resistance changes and inductance variation due to magnetic saturation should be studied, since the NMHE prediction model assumes fixed parameters. To achieve fully sensorless operation from standstill, initial rotor position detection methods based on high-frequency signal injection or short voltage pulse injection can be explored to eliminate Hall sensor dependency during startup. Additionally, Integrating the NMHE with online parameter estimation techniques would also make the sensorless drive more adaptive to varying motor and load conditions.

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Data availability Data will be available from the corresponding author by proper request.

Code availability The MATLAB/Simulink implementation of the NMHE estimator and the PSO, DEA, and GWO PI-tuning routines used in this study is openly available in the Zenodo repository, DOI [to be inserted on acceptance].

Declarations

Competing interests The authors declare no competing interests.

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